

PROJECT REPORT

Personal Portfolio Website Using Flask

1. Introduction

The Personal Portfolio Website is a full-stack web application developed using Flask, HTML, and CSS. The main objective of this project is to create a responsive and interactive website that showcases personal details, technical skills, projects, and contact information.

This project helps beginners understand web development concepts including frontend design, backend integration, routing, and deployment basics using Python Flask.

2. Objectives

- To build a responsive portfolio website
- To learn Flask framework basics
- To understand frontend and backend integration
- To display projects and skills professionally
- To host and access the website online using ngrok

3. Technologies Used

| Technology | Purpose |

|---|---|

| Python | Backend programming |

| Flask | Web framework |

| HTML | Structure of webpage |

| CSS | Styling and design |

| Google Colab | Development environment |

| Flask-Ngrok | Public URL generation |

4. System Requirements

Hardware Requirements

- Laptop/Desktop/Mobile
- Minimum 4GB RAM
- Internet Connection

Software Requirements

- Google Colab
- Python 3
- Flask Library
- Web Browser

5. Project Structure

personal-portfolio-website/

├─ app.py

├─ templates/

| └─ index.html

├─ static/

| └─ style.css

└─ README.md

6. Working of the Project

1. Flask server starts using Python.
2. The browser sends a request to the Flask application.
3. Flask routes the request to the correct webpage.
4. HTML displays the webpage content.
5. CSS styles the webpage.
6. Ngrok generates a public link for accessing the website online.

7. Features

- Responsive portfolio design
- Skills section
- Projects section
- Contact details
- Flask routing
- Public URL access
- Beginner-friendly interface

8. Advantages

- Simple and easy to develop
- Lightweight framework
- Professional online presence
- Easy customization
- Can be deployed online

9. Limitations

- Basic frontend design
- No database integration
- No authentication system
- Static project information

10. Future Enhancements

- Add database connectivity
- Add user authentication
- Add downloadable resume
- Add contact form
- Deploy using cloud platforms
- Add animations and dark mode

11. Conclusion

The Personal Portfolio Website project successfully demonstrates the integration of Flask with HTML and CSS for creating a professional portfolio application. This project improves understanding of full-stack web development and helps in building practical development skills.

12. References

- Python Documentation
- Flask Official Documentation
- Google Colab
- HTML & CSS Tutorials

13. Author

Name: Bhavani